Purine Degradation & Gout

- The end product of purine metabolism in humans is uric acid.
- The nucleotide monophosphates (AMP, IMP & GMP) are converted to their respective nucleoside forms (adenosine, inosine & guanosine) by the action of nucleotidase.
- The amino group, either from AMP or adenosine, can be removed to produce IMP or ionosine.

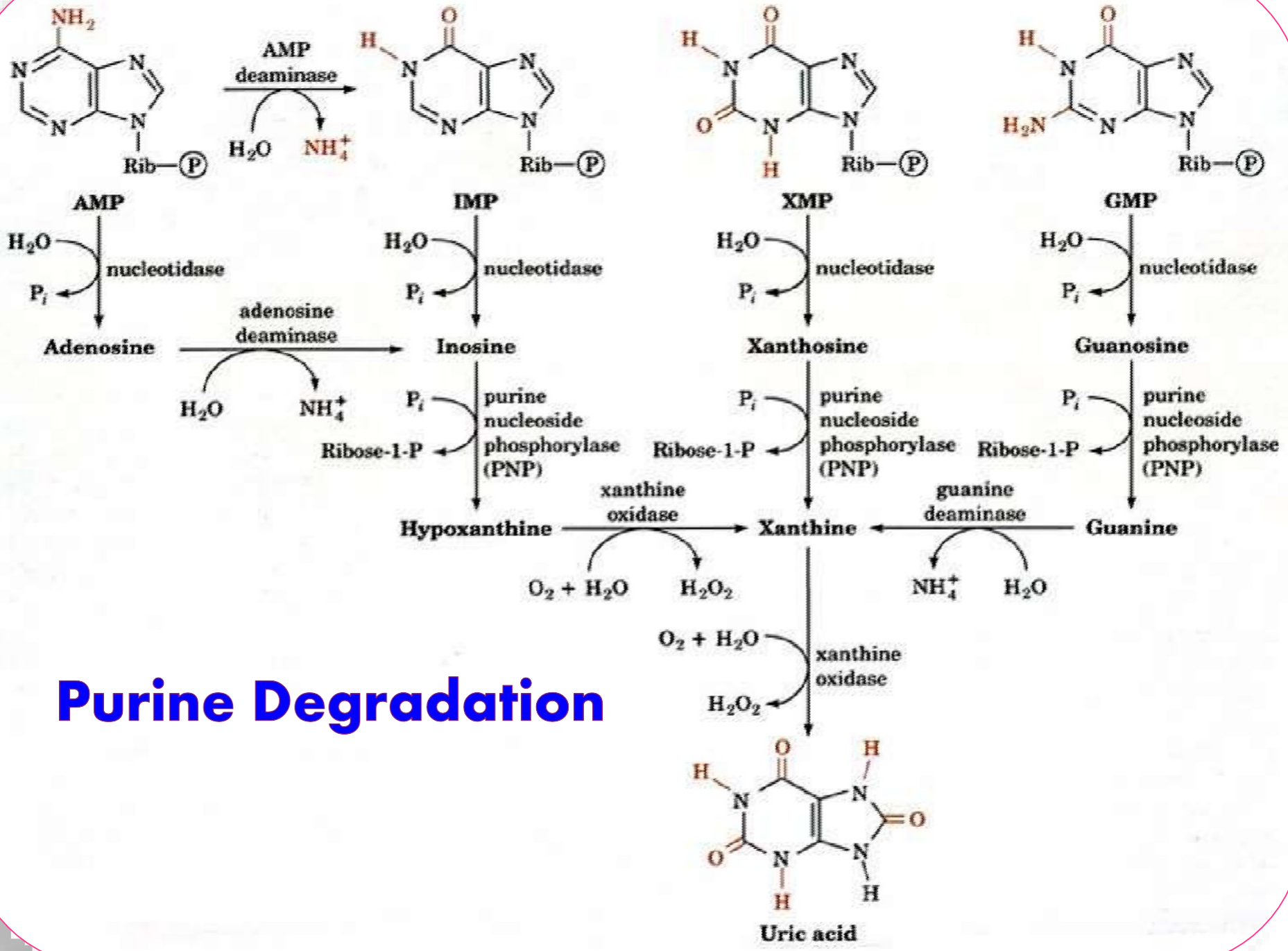
- Inosine & guanosine are converted to hypoxanthine & guanine (purine bases) by purine nucleoside phosphorylase.
- Adenosine is not degraded by this enzyme, it has to be converted to inosine.
- Guanine undergoes deamination by guanase to form xanthine.

- Xanthine oxidase converts hypoxanthine to xanthine & xanthine to uric acid.
- This enzyme contains FAD, molybdenum & iron.
- It is exclusively found in liver & small intestine.
- Xanthine oxidase liberates H_2O_2 which is harmful to the tissues.
- Catalase cleaves H_2O_2 to H_2O & O_2.

Uric acid (2,6,8-trioxypurine)

- It is final excretory product of purine metabolism in humans.
- Uric acid can serve as an important antioxidant by getting itself converted (non-enzymatically) to allantoin.
- It is believed that the antioxidant role of ascorbic acid in primates is replaced by uric acid.

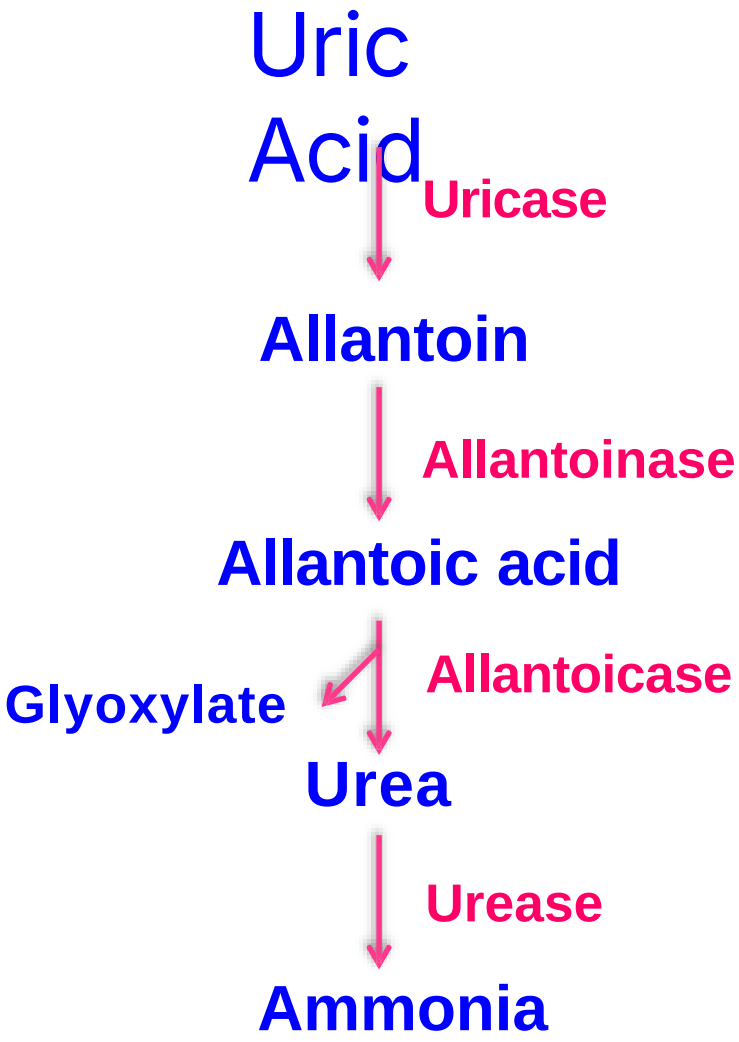
Purine Degradation



Degradation of Uric acid in animals other than man

- Most animals (other than primates) oxidize uric acid by the enzyme uricase to allantoin, where the purine ring is cleaved.
- Allantoin is then converted to allantoic acid & excreted in some fishes.
- Further degradation of allantoic acid may occur to produce urea (in amphibians, most fishes & some molluscs) & to ammonia (in marine invertebrates).

Degradation of Uric acid in animals other than man



Disorders of purine metabolism

- Normal blood level of uric acid.
- Females: 2 - 5 mg/dl.
- Males: 3 - 7 mg/dl.
- The daily excretion varies from 500-700 mg.
- Nucleic acid content is more in non-vegetarian diet.
- Uric acid is sparingly soluble in water.

- Hyperuricemia refers to an elevation in the serum uric acid concentration.
- This is sometimes associated with increased uric acid excretion (uricosuria).

Gout

- It is a metabolic disease associated with overproduction of uric acid.
- At the physiological pH, uric acid is found in a more soluble form as sodium urate.
- In severe hyperuricemia, crystals of sodium urate get deposited in the soft tissues, particularly in the joints.
- Such deposits are commonly known as tophi.

- This causes inflammation in the joints resulting in a painful gouty arthritis.
- Sodium urate or uric acid may also precipitate in kidneys & ureters that results in renal damage & stone formation.
- Gout is of two types:
- Primary
- Secondary




Uric Acid
Build up
Inflamed
Joint
Sharp Needle Like
Uric Acid Crystals

Primary Gout

- It is due to overproduction of uric acid.
- This is mostly related to increased synthesis of purine nucleotides.
- About 10% of cases of primary gout are idiopathic.
- Incidence of primary gout is about 3 per 1,000.
- Mostly affecting males.

Causes of Primary Gout

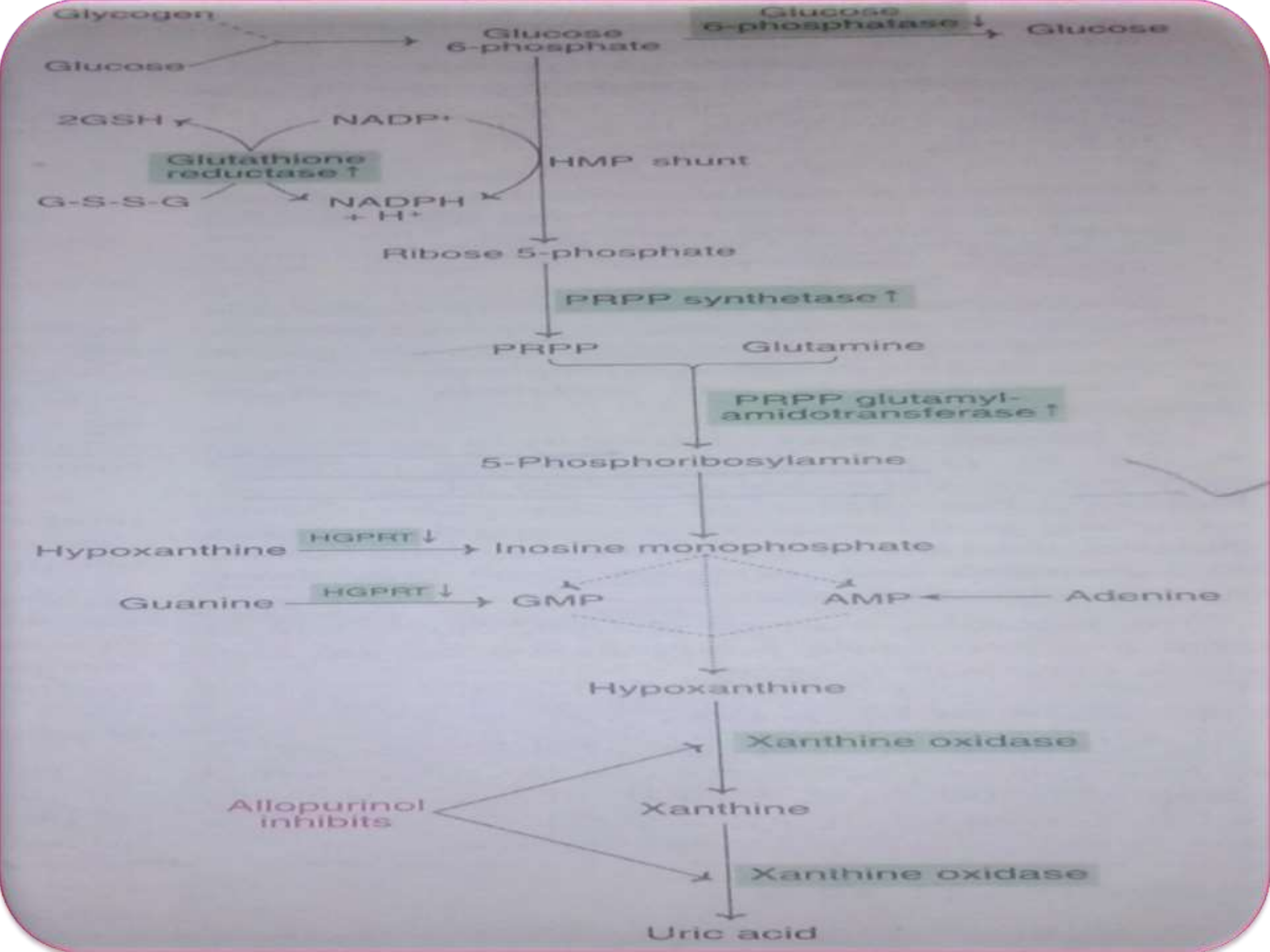
- **PRPP synthetase:**
- **In normal circumstances, PRPP synthetase is under feedback control by purine nucleotides (ADP & GDP).**
- **A variant forms of PRPP synthetase-which are not subjected to feedback regulation-leads to the increased production of purines.**

- **PRPP glutamylamidotransferase:**
- **The lack of feedback control of this enzyme by purine nucleotides also leads to their elevated synthesis.**
- **Deficiency of enzymes of salvage pathway:**
- **HGPRT enzyme of purine salvage pathway & its defect causes Lesch-Nyhan syndrome.**
- **It is associated with increased synthesis of purine nucleotides.**

- **Firstly, decreased utilization of purines (hypoxanthine & guanine) by salvage pathway.**
- **Resulting in accumulation & diversion of PRPP for purine nucleotides.**
- **Secondly, the defect in salvage pathway leads to decreased levels of IMP & GMP causing impairment in the tightly controlled feedback regulation of their production.**

- **Glucose-6-phosphatase deficiency:**
- **This condition is known as von Gierke's disease (glycogen storage disease, type I)**
- **When this enzyme is deficient, glucose-6-phosphate cannot be converted to glucose.**
- **More glucose is channelled into the HMP shunt, resulting in increased availability of ribose-5-phosphate.**

- This leads to increased formation of PRPP & purine over production.
- Elevation of glutathione reductase:
- Increased glutathione reductase generates more $NADP^+$ which is utilized by HMP shunt.
- This causes increased ribose 5-phosphate & PRPP synthesis

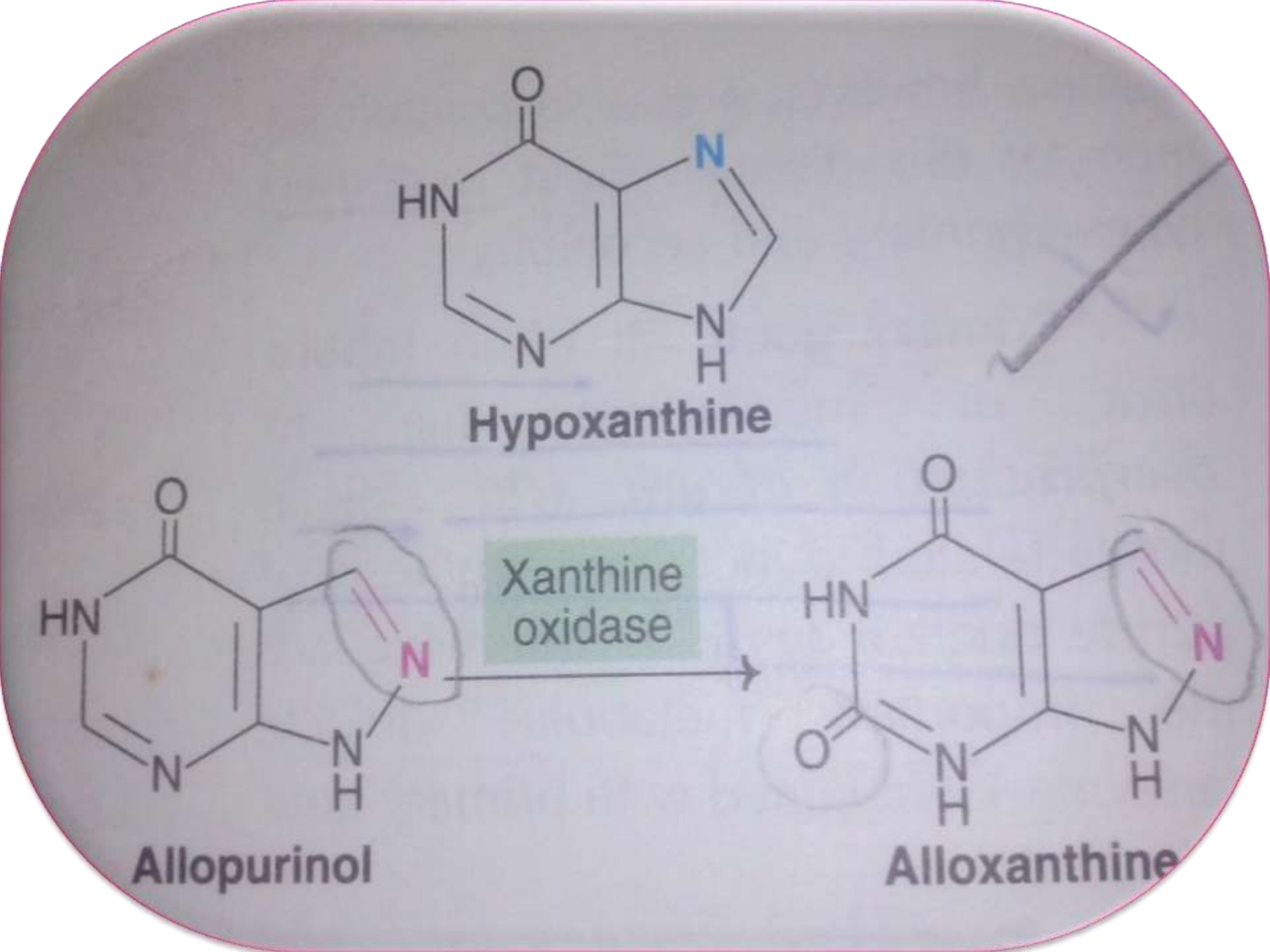

Glycogen
Glucose
Glucose 6-phosphate
Glucose 6-phosphatase ↓
Glucose
2GSH
NADP+
Glutathione reductase ↑
G-S-S-G
NADPH + H+
HMP shunt
Ribose 5-phosphate
PRPP synthetase ↑
PRPP
Glutamine
PRPP glutamyl-amidotransferase ↑
5-Phosphoribosylamine
Hypoxanthine
HGPRT ↓
Inosine monophosphate
Guanine
HGPRT ↓
GMP
AMP
Adenine
Hypoxanthine
Xanthine oxidase
Allopurinol inhibits
Xanthine
Xanthine oxidase
Uric acid

Secondary Gout

- It is due to various diseases causing increased synthesis or decreased excretion of uric acid.
- Increased degradation of nucleic acids is observed in various cancers (leukemias, polycythemia, lymphomas, etc.) psoriasis & increased tissue breakdown (trauma, starvation etc.) & impairment in renal function cause accumulation of uric acid, lead to gout.

Treatment of gout

- The drug, allopurinol is used for primary gout.
- It is a structural analog of hypoxanthine that competitively inhibits enzyme xanthine oxidase.
- Allopurinol is oxidized to alloxanthine by xanthine oxidase.
- Alloxanthine, is a more effective inhibitor of xanthine oxidase & is referred to as suicide inhibition.


Hypoxanthine
Xanthine oxidase
Allopurinol
Alloxanthine

Xanthine oxidase

Allopurinol → **Alloxanthine**

- Inhibition of xanthine oxidase by allopurinol leads to the accumulation of hypoxanthine & xanthine, water soluble & easily excreted.
- Restriction in dietary intake of purines & alcohol is advised.
- Consumption of plenty of water is useful.
- The anti-inflammatory drug colchicine is used for the treatment of gouty arthritis.

Pseudogout

- The clinical manifestations of pseudogout are similar to gout.
- It is caused by the deposition of calcium pyrophosphate crystals in joints.
- Serum uric acid concentration is normal in pseudogout.

Lesch-Nyhan syndrome

- It is due to the deficiency of hypoxanthine-guanine phosphoribosyltransferase (HGPRT), an enzyme of purine salvage pathway.
- Lesch-Nyhan syndrome is a sex-linked metabolic disorder since the structural gene for HGPRT is located on the X-chromosome.

- It affects only the males.
- Characterized by excessive uric acid production (Gouty arthritis) & neurological abnormalities.
- Mental retardation, aggressive behavior, learning disability etc.
- The patients of this disorder have an irresistible urge to bite their fingers & lips, often causing self-mutilation.

Lesch-Nyhan syndrome



- **Over production of uric acid in Lescn-Nyhan syndrome:**
- **HGPRT deficiency results in the accumulation of PRPP & decrease in GMP & IMP, ultimately leading to increased synthesis & degradation of purines.**

Hypouricemia

- Adenosine Deaminase (ADA) deficiency.
- Deficiency of ADA causes severe combined immunodeficiency (SCID) involving T-cell & usually B-cell dysfunction.
- ADA deficiency results in the accumulation of dATP which is an inhibitor of ribonucleotide reductase & DNA synthesis & cell replication.

- The deficiency of purine nucleotide phosphorylase is associated with impairment of T-cell function but has no effect on B-cell function.
- Uric acid synthesis is decreased & the tissue levels of purine nucleosides & nucleotides are higher.
- dGTP inhibits the development of normal T-cells.

Hypouricemia

- Decreased uric acid levels in the serum (< 2 mg/dl) called as hypouricemia.
- This is mostly associated with a rare genetic defect in the enzyme xanthine oxidase.
- It leads to the increased excretion of xanthine & hypoxanthine.
- Xanthinuria frequently causes the formation of xanthine stones in the urinary tract.

References

- Lippincott
- **Chatterjea**